ENGR 689 - Multimodal LLM Agents (Fall 2026): Quiz 1

Don't forget to write your name on the paper you submit!
Each question has only one correct option.
8 questions **Duration:** 25 Minutes

Q1 (1%) Which statement best explains why BERT substantially changed the traditional NLP workflow?

A. It eliminated the need to evaluate models on individual tasks.
B. It allowed a single fixed output format to solve every NLP task without adaptation.
C. It learned reusable language patterns and general knowledge from large text corpora and could then be adapted using less task-specific training data.
D. It guaranteed that models could follow any previously unseen instruction in a zero-shot setting.

Q2 (1%) Which statement best explains how a single GPT-3 model can be applied to many different NLP tasks?

A. GPT-3 trains a separate set of model parameters for every downstream task.
B. GPT-3 is pre-trained using next-token prediction, and many NLP tasks can be formulated as text continuation within a prompt.
C. GPT-3 updates its model parameters whenever examples are included in the prompt.
D. GPT-3 solves every task by first retrieving the answer from an external corpus.

Q3 (1%) Which prompt formulation most directly casts a paper-classification task as next-token prediction?

A. Provide the paper title followed by "`Category:`" and ask the model to generate the category name.
B. Retrieve the most similar paper and copy its abstract.
C. Ask a human annotator to assign a category to every test paper.
D. Add a task-specific classification head that directly outputs one of several fixed category IDs.

Q4 (1%) At inference time, a model is given three input-output examples in its prompt and then correctly handles a new input without updating its parameters. What does this illustrate?

A. Pre-training
B. Supervised fine-tuning
C. Retrieval-augmented generation
D. In-context learning

Q5 (2%) A system includes worked-out reasoning steps in its demonstration examples. At test time, it generates eight different reasoning paths and selects the most frequent final answer. Which two techniques is it combining?

A. Instruction tuning and retrieval-augmented generation

B. Chain-of-thought prompting and self-consistency

C. Pre-training and supervised fine-tuning

D. Zero-shot prompting and reinforcement learning from human feedback

Q6 (1%) The goal is to improve an LLM's ability to follow previously unseen task instructions in a zero-shot setting. Which type of training data would most directly support this goal?

A. A very large number of examples from a single sentiment-classification task

B. Additional unlabeled web text without any task instructions

C. Examples from many different tasks, each represented as an instruction, input, and output

D. Human preference comparisons collected for only one narrow task

Q7 (2%) Suppose I want to use an LLM to recommend a movie based on a user's viewing history. One possible prompt is:

```
A user has watched the following movies, in this order:

Harry Potter and the Sorcerer's Stone
-> Harry Potter and the Chamber of Secrets
-> Harry Potter and the Prisoner of Azkaban

Based on this viewing history, which movie should be recommended next?
```

Which statement about using this prompt is correct?

A. This prompt uses chain-of-thought prompting because the movies in the user's viewing history form a sequence (i.e., "chain").

B. GPT-3 should reliably follow this zero-shot instruction because web-scale next-token pre-training directly trains it to follow diverse task instructions.

C. InstructGPT updates its model parameters using the three previously watched movies before generating a recommendation.

D. Compared with the original GPT-3, InstructGPT is more likely to follow this prompt because instruction tuning trains it to follow diverse natural-language task instructions.

Q8 (1%) An LLM must answer an up-to-date question whose supporting evidence is distributed across several web pages. Which approach best represents agentic retrieval-augmented generation (RAG)?

A. Generate the answer entirely from the model's existing parameters.

B. Fine-tune the model on one newly collected fact before answering each query.

C. Retrieve one document using a fixed query, regardless of whether the retrieved document is sufficient.

D. Let the model decide when to search, reformulate its queries, perform multiple rounds of retrieval, and use the retrieved evidence to answer the question.